

Senior Software Engineer

Summary

- Sr. Software Engineer with 11+ years of experience focused on the scientific projects
- Proficient in C, Go, Python, Matlab [ordered by preference]
- Linux kernel development & exploitation (6 y), pentest (OSCP, OSCE certified)
- 4 years experience in wireless RF projects: GSM (osmocom), U-SIM (RAM/RFM), LTE(-A/Pro) decoding L1-L3
- Self-studying the quantum information & computation

Experience [inDrive](#), full-time remote

Senior SW Engineer II (Feb 2024 - present)

- Backend engineering on Go, Python for the ride-hailing, recommended price, bidding, geo-spatial related tasks

[TryHackMe](#), contract

System SW Engineer (March 2024 - present)

- eBPF based C programs development for the players' actions monitoring to support their pentesting journey with "Echo" AI assistant.
- Developed in Go a lightweight userspace program to monitor fs, network, Linux LPE attempt tracking, execve(), uprobes (bash, msfconsole symbols), PID/directory hiding, kprobes etc.

Freelance, contract

RF SW Engineer (April 2022 - June 2023)

- ["ORM" project](#) - SW + HW with C (2K LOC), Matlab (8K LOC) to decode radio to human-readable format
- Signal capture (1 ms real-time) to eCPRI format via bladeRF driver, Xilinx RFSoc architecture research
- GNURadio programming with C++ for spectrogram analysis (1 ms subframe 2D analysis + OFDM power grid)

[Spectre Security Group](#), Kazakhstan, on-site

CTO, Co-founder (Sept 2020 - Aug 2022)

- Team of 6 top-notch security experts who are the most famous in Kazakhstan and Standoff CTF winners
- OTA-SMS (with SIMTracer), A-GPS spoofing, vulnerabilities research (Motorola C123 + osmocom, YateBTS)
- U-SIM filesystem storage for encrypted 2FA and Java applet development for RFM

[TSARKA](#), Astana, Kazakhstan, on-site

Embedded SW Engineer (Nov 2019 - Sept 2020)

- CVE-2019-2215: N-day Android binder UAF exploit development on Samsung device
- Linux driver modification for I2C serial comm. on NXP devboards. (SoC-to-SoC) on Yocto built OS
- Radio base station programming on the ESP-32. Working with the smart home IoT protocols

[Google](#), Sunnyvale, CA, USA, on-site, contract

Application Engineer (Nov 2018 - Nov 2019)

- DevOps for Google Search Appliance project with Python scripts for network Linux installation via PXE
- Bare metal servers' (< 500) maintenance and utilization, Linux kernel OOMKiller configuration
- systemd CVE-2018-16865, CVE-2018-16864, CVE-2018-16866 research & POC development in C

[EPAM Systems](#), Kazakhstan, on-site

Senior SW Engineer (June 2013 - Nov 2018)

- Participated in a variety of enterprise projects. Mostly gained a lot of experience with customer comm. per ITIL process. Understand solution & system architecture. Func/non-func requirements' and SLAs fulfillment
- k8s clusters' administration, architecture on Azure, AWS. Developed a fully automated CI/CD pipeline on Jenkins from scratch on Groovy. Configured multiple ENV for each k8s clusters, automated with Helm templates

[Genesys Telecommunications](#), remote, contract

SW Engineer (Nov 2014 - Jan 2018)

- Developed a customer-experience platform with NLP analytical features. Customer contacts to the company, depending on DTMF, customer profile and other input, is routed to the best agent fit. Tech: MongoDB, Python, NLP, sklearn, JS visualization of various analytical 2D charts

Education

Bachelor of Science in Mechanical Engineering, KSTU, Kazakhstan (2011-2015), GPA: 3.46

Misc

IBM - Basics of Quantum Information (2026), OSCP (2019), OSCE (2020), Autopsy forensic, Linux kernel development course (LFD420) by Linux Foundation. Lecturing graduate students (max. 67 listeners) & technical leadership in small teams (up to 5 engineers)